

PaperPath

I want to build an AI-powered educational platform that makes research papers easier to discover, understand, and learn from, especially for beginners and students. The core problem being solved is that current research platforms such as Google Scholar and arXiv are designed mainly for researchers and advanced users, which makes research papers intimidating and difficult for school students or early college learners. There is currently no structured platform that helps beginners gradually enter the world of research papers in a guided and understandable way.

The idea is to create a platform that works similarly to a “LeetCode for research papers.” Users will be able to discover research papers categorized by difficulty levels such as beginner (Class 9–10), intermediate (Class 11–12), and advanced (college level). The platform will include community-driven ratings for difficulty, clarity, usefulness, and learning value. The goal is not just to store papers, but to turn research reading into a structured learning journey.

The platform will also include AI-powered features. Users can upload or open a research paper and get simplified explanations, summaries, key concepts, prerequisites, and beginner-friendly interpretations. The AI system should be capable of explaining difficult paragraphs in simpler language depending on the learner’s level. For example, a user could ask: “Explain this paragraph like I’m in class 10,” and the AI would respond in simplified language with analogies and easy explanations.

The product is essentially a combination of:

- * An educational platform
- * A research discovery system
- * A community rating platform
- * An AI explanation engine
- * A future recommendation and learning-path system

The MVP (minimum viable product) should remain small and focused. The first version should only include:

- * User authentication
- * Research paper listing
- * Difficulty tagging
- * Ratings and reviews
- * AI-generated summaries and explanations
- * Search and filtering

The current recommended tech stack is:

- * Frontend: Next.js + Tailwind CSS
- * Backend: FastAPI
- * Database: PostgreSQL

- * Authentication: JWT
- * AI Development: Ollama locally during development
- * AI Production: Gemini API or OpenAI API for easier deployment
- * Vector Database: ChromaDB
- * PDF Processing: PyMuPDF or pdfplumber
- * AI Framework: LangChain
- * Deployment: Vercel (frontend) + Railway/Render (backend and DB)

The AI system should use RAG (Retrieval Augmented Generation) because research papers are large and complex. The workflow is:

1. Upload or fetch PDF
2. Extract text from PDF
3. Clean and preprocess text
4. Split text into chunks
5. Generate embeddings
6. Store embeddings in a vector database
7. Retrieve relevant chunks based on user queries
8. Send relevant context to the LLM
9. Generate beginner-friendly explanations and summaries

The most important AI concepts to learn are:

- * Prompt engineering
- * Structured prompting
- * Context engineering
- * RAG pipelines
- * Embeddings
- * Vector databases
- * PDF processing
- * LLM workflow design
- * AI evaluation and hallucination reduction

The focus should NOT be on training custom LLMs or deep machine learning theory. The project is more about AI systems engineering and intelligent workflow design than traditional ML research.

Current strengths of the developer:

- * Strong DSA background (700+ LeetCode problems solved)
- * MERN stack knowledge
- * React experience
- * Authentication systems
- * API integration
- * Basic Generative AI experience
- * Git/GitHub familiarity

Main areas to improve:

- * System design

- * PostgreSQL and relational database modeling
- * FastAPI
- * RAG architecture
- * Prompt engineering
- * Product thinking and MVP prioritization
- * AI workflow design

The project has strong startup and hackathon potential because it solves a real educational problem. The main differentiation is that the platform is designed specifically for beginners and progressive learning rather than only for researchers. The long-term vision is to make research papers accessible, structured, and understandable for everyone, starting from school students to advanced learners.